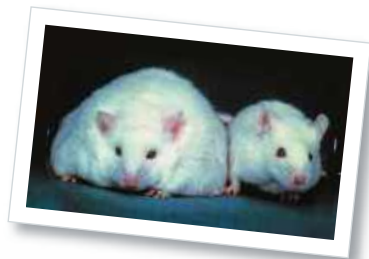


Interpreting Data from an Experiment with Genetic Mutants

What Are the Roles of the *ob* and *db* Genes in Appetite Regulation? A mutation that disrupts a physiological process is often used to study the normal function of the mutated gene. Ideally, researchers use a standard set of conditions and compare animals that differ genetically only in whether a particular gene is mutant (nonfunctional) or wild-type (normal). In this way, a difference in phenotype, the physiological property being measured, can be attributed to a difference in genotype, the presence or absence of the mutation. To study the role of specific genes in regulating appetite, researchers used laboratory animals with known mutations in those genes.

Mice in which recessive mutations inactivate both copies of either the *ob* gene or the *db* gene eat voraciously and grow much more massive than wild-type mice.

▶ The mouse on the right is wild-type, whereas the obese mouse on the left has an inactivating mutation in both copies of the *ob* gene.



One hypothesis for the normal role of the *ob* and *db* genes is that they participate in a hormone pathway that suppresses appetite when caloric intake is sufficient. Before setting out to isolate the potential hormone, researchers explored this hypothesis genetically.

How the Experiment Was Done The researchers measured the mass of young mice of various genotypes (the “subjects”) and surgically linked the circulatory system of each one to that of another mouse. This procedure ensured that any factor circulating in the bloodstream of either mouse would be transferred to

the other in the pair. After eight weeks, they again measured the mass of each subject mouse.

Data from the Experiment

	Genotype Pairing (red type indicates mutant genes)		Average Change in Body Mass of Subject (g)
	Subject	Paired with	
(a)	<i>ob</i> ⁺ / <i>ob</i> ⁺ , <i>db</i> ⁺ / <i>db</i> ⁺	<i>ob</i> ⁺ / <i>ob</i> ⁺ , <i>db</i> ⁺ / <i>db</i> ⁺	8.3
(b)	<i>ob</i> / <i>ob</i> , <i>db</i> ⁺ / <i>db</i> ⁺	<i>ob</i> / <i>ob</i> , <i>db</i> ⁺ / <i>db</i> ⁺	38.7
(c)	<i>ob</i> / <i>ob</i> , <i>db</i> ⁺ / <i>db</i> ⁺	<i>ob</i> ⁺ / <i>ob</i> ⁺ , <i>db</i> ⁺ / <i>db</i> ⁺	8.2
(d)	<i>ob</i> / <i>ob</i> , <i>db</i> ⁺ / <i>db</i> ⁺	<i>ob</i> ⁺ / <i>ob</i> ⁺ , <i>db</i> / <i>db</i>	−14.9*

*Due to pronounced weight loss and weakening, subjects in this pairing were remeasured after less than eight weeks.

Data from D. L. Coleman, Effects of parabiosis of obese mice with diabetes and normal mice, *Diabetologia* 9:294–298 (1973).

INTERPRET THE DATA

- First, practice reading the genotype information given in the data table. For example, pairing (a) joined two mice that each had the wild-type version of both genes. Describe the two mice in pairing (b), pairing (c), and pairing (d). Explain how each pairing contributed to the experimental design.
- Compare the results observed for pairing (a) and pairing (b) in terms of phenotype. If the results had been identical for these two pairings, what would that outcome have implied about the experimental design?
- Compare the results observed for pairing (c) to those observed for pairing (b). Based on these results, does the *ob*⁺ gene product appear to promote or suppress appetite? Explain your answer.
- Describe the results observed for pairing (d). Note how these results differ from those for pairing (b). Suggest a hypothesis to explain this difference. How could you test your hypothesis using the kinds of mice in this study?

➔ **Instructors:** A version of this Scientific Skills Exercise can be assigned in **Mastering Biology**.

stomach wall, triggers feelings of hunger before meals. In contrast, both insulin and *PYY*, a hormone secreted by the small intestine after meals, suppress appetite. *Leptin*, a hormone produced by adipose (fat) tissue, also suppresses appetite and appears to play a major role in regulating body fat levels. In the **Scientific Skills Exercise**, you’ll interpret data from an experiment studying genes that affect leptin production and function in mice.

Obtaining food, digesting it, and absorbing nutrients are part of the larger story of how animals fuel their activities. Provisioning the body also involves circulating nutrients, and using nutrients for metabolism requires exchanging respiratory gases with the environment. These processes and the adaptations that facilitate such distribution and exchange are the focus of Chapter 42.

CONCEPT CHECK 41.5

- Explain how people can become obese even if their intake of dietary fat is relatively low compared with carbohydrate intake.
- WHAT IF?** Suppose you were studying two groups of people who are obese and have genetic abnormalities in the leptin pathway. In one group, the leptin levels are abnormally high; in the other group, they are abnormally low. How would each group’s leptin levels change if they ate a low-calorie diet for an extended period? Explain.
- WHAT IF?** An insulinoma is a cancerous mass of pancreatic beta cells that secrete insulin but do not respond to feedback mechanisms. How you would expect an insulinoma to affect blood glucose levels and liver activity?

For suggested answers, see Appendix A.